Reg. No. | | | | | | | | | | |

[18MTAVS201]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

NAVIGATION AND COMMUNICATION SYSTEMS

Time: 03 Hours **Maximum Marks: 100**

Instructions:
1. Answer Five Full Questions.
2. Question numbers one and two are compulsory.
3. Draw neat sketches wherever necessary.

PART - A

1. a. Define the following with suitable diagrams **10**
   - i. Altitude
   - ii. Longitude
   - iii. Geodesy, Compare the salient features of Longitude and Latitude

   b. Explain the operation of DME system used in aircraft Navigation and what are its limitations. **10**

2. a. Explain the operation of a Super heterodyne receiver with multiple IF stage with the help of a block diagram. **10**

   b. Explain the Pulse code modulation technique with a block diagram and its application. **10**

3. a. Explain the mathematical model to estimate the distance in a satellite navigation system. **10**

   b. Explain the operation of mechanical and MEMS based accelerometers for aircraft application. **10**

OR

4. a. Explain with neat diagram the operation of Automatic Direction Finder(ADF) used in Radio Navigation **10**

   b. Explain the operation of GPS receiver with neat diagram and its applications **10**

PART - B

5. a. Explain the operation of Differential GPS. **08**

   b. Explain the integration of GPS/INS with Kalman filter for feed forward aided INS and feedback aided INS with suitable block diagram **08**

   c. Explain the differences between GPS and GLONASS. **04**

OR

6. a. Explain sampling theorem and quantization error. **04**

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	b.	A modulated carrier is viewed and measured on an oscilloscope as follows. The maximum /crest voltage of an modulated carrier wave is 35Volts.The minimum/trough voltage is 5Volts.What is the percentage of modulation. Estimate the depth of modulation when the crest and trough voltage of the resultant modulated carrier wave is changed to 60 % .	**10**
	c.	Explain the ACARS system used in the aircraft.	**06**
7.	a.	Explain AM and FM modulation process and its characteristics.	**10**
	b.	Explain the image frequency and image frequency rejection and how it can be improved in a Super heterodyne receiver.	**06**
	c.	Explain the working of Radomes and its application.	**04**
		OR	
8.	a.	Explain the operation of a dipole antenna ant its radiation pattern with a neat diagram.	**10**
	b.	Explain the operation of antenna's used in satellite communication .	**10**